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# DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

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## ABSTRACT

AI weather models are often reported to underestimate extremes, but most evidence concerns deterministic regression models verified against reanalysis. We evaluate twelve physical and AI forecast models against ECMWF IFS using ten months of European station observations. The evaluation covers 10 m wind, 2 m temperature, solar radiation, and precipitation within regimes defined from a fixed ERA5 1991–2020 climatology. We find no uniform AI-specific deficit in the tails. Several AI models remain more accurate than IFS under extreme conditions, while others deteriorate markedly; comparable variation occurs among physical models. Every model nevertheless exhibits a common conditional-error pattern, overpredicting low observations and underpredicting high observations. Attenuation of extreme values therefore does not imply a uniform loss of relative skill: tail performance depends on the model, variable, and evaluation setting rather than on whether the forecast is produced by AI or physical numerical modelling.

## 1 INTRODUCTION

Data-driven weather models have achieved strong performance on standard forecast metrics, often matching or surpassing operational numerical weather prediction (NWP) models (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). However, many of these models produce deterministic forecasts and are trained using mean-squared error. These models tend to regress towards the conditional mean, smoothing small-scale variability and underestimating extremes, as reported for peak winds and record-breaking events (Olivetti & Messori, 2024a; Bonavita, 2024; Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

Most existing evidence concerns deterministic regression models evaluated against reanalysis. Probabilistically trained models, generative ensembles, and high-resolution regional models have received less attention, particularly when verified against weather stations, where errors can be substantially larger (Jin et al., 2024; Gupta et al., 2025; Schlueter & Wagner, 2025). Generative models are particularly relevant because they represent a distribution of possible outcomes rather than a single conditional estimate, and recent work has shown promising performance for weather extremes and high-resolution regional forecasting (Price et al., 2025; Pathak et al., 2026; Molinaro et al., 2024).

A second limitation is that underestimation is often attributed specifically to AI, even though also NWP models may underestimate the same observations. An AI model can therefore underestimate an rare event while remaining closer to it than an NWP forecast. The relevant question is not whether AI models underestimate extremes, but whether they are less accurate than NWP when extremes occur.

We test this using thirteen forecast models, including ECMWF IFS as the reference. Forecasts are evaluated over ten months against European synoptic, solar-radiation, and rain-gauge observations. The analysis covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly precipitation at lead times up to 48 h. We evaluate extremes in two complementary ways: first, by stratifying forecasts according to locally defined climatological regimes; and second, through externally documented record-breaking temperature events and major European windstorms.

Our main contributions are:

- We extend station-based evaluation of weather extremes to deterministic regression models, a CRPS-trained global ensemble, and high-resolution regional generative models.
- We distinguish conditional forecast errors from relative accuracy, testing whether models that lie below high observations are also less accurate than IFS under the same conditions.
- We evaluate performance across four surface variables, forecast leads, countries, and model classes, and complement the climatological analysis with source-defined temperature-record events, major windstorms, and categorical verification of heavy precipitation.

## 2 METHODS

We describe the forecast models and station observations, then introduce the climatological regimes, bias correction, and evaluation protocol.

### 2.1 MODELS

Let  $u(t)$  denote the atmospheric state at time  $t$ . Weather forecasting amounts to approximating its evolution from an initial state  $u(t_0)$ ,

$$\hat{u}_m(t_0 + \tau) = \mathcal{F}_m^\tau(u(t_0)), \quad (1)$$

where  $m$  indexes the forecast model and  $\tau$  is the forecast lead time. NWP models approximate  $\mathcal{F}_m^\tau$  through numerical integration of the governing equations on a discrete grid, whereas AI models learn this mapping from historical atmospheric states. Forecasts may be deterministic, consisting of a single state, or probabilistic, consisting of an ensemble,

$$\left\{ \hat{u}_m^{(k)}(t_0 + \tau) \right\}_{k=1}^K.$$

We evaluate twelve forecast models against the high-resolution deterministic ECMWF IFS forecast, which serves as the reference (Table S1; Appendix A.1). The comparison spans NWP and AI models, global and regional domains, and deterministic and ensemble forecasts.

The NWP models are ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models include ECMWF AIFS (Lang et al., 2024a), Microsoft Aurora (Bodnar et al., 2025), and Jua EPT-2 Reasoning (Molinaro et al., 2025), all global deterministic regression models. The global ensembles are ECMWF AIFS ENS (Lang et al., 2024b), a 51-member ensemble trained with a continuous ranked probability score objective and learned noise injection, and Jua EPT-2e (Molinaro et al., 2025), a perturbation-based ensemble. The regional models are Jua EPT-2 HRRR (Siegenheim, 2025) and EPT-2.1 Europa (Jua AI, 2026), generative ensembles based on the methodology of Molinaro et al. (2026; 2024), and EPT-2.1 Helios (Molinaro, 2026), a solar-specialised regional model.

Models are evaluated only for variables with compatible outputs. Precipitation is available from the NWP models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS, AIFS ENS, and Aurora are restricted to wind and temperature because their radiation and precipitation outputs cannot be matched to the hourly verification data.

### 2.2 STATION OBSERVATIONS

Forecasts are verified against three complementary in situ datasets: operational surface observations for 10 m wind and 2 m temperature, solar-radiation observations for hourly global horizontal irradiance, and observations from 4,005 rain gauges for hourly precipitation. Specific contributing networks and data providers are listed in Appendix A.

### 2.3 OBSERVED-VALUE REGIMES

Regimes are defined independently of the evaluation period using a fixed ERA5 1991–2020 climatology (Hersbach et al., 2020). ERA5 fields are bilinearly interpolated from the  $0.25^\circ$  grid to each

station. For each station, calendar day, and UTC hour, the reference sample comprises values from the same UTC hour within a cyclic  $\pm 15$ -day window over the 30-year period, yielding 930 values before variable-specific filtering.

Percentile thresholds are computed separately from each reference sample. Wind and temperature use all reference values, whereas solar radiation uses positive ERA5 GHI values only. For precipitation, values below 0.1 mm define a separate dry regime, and P50, P75, and P95 are computed from wet reference hours only. Complete definitions and meteorological labels are given in Tables S2 and S3.

This local-climatology construction follows established practice for climate extremes (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander, 2013). Each verifying observation  $y$  is assigned to a regime by comparison with the thresholds for its station, calendar day, and UTC hour.

## 2.4 BIAS CORRECTION

Persistent model-wide biases can dominate forecast errors and obscure how performance changes across observed-value regimes. To reduce their influence, we apply a causal rolling bias correction before computing skill. For wind, temperature, and daytime solar radiation, the correction is additive. Let  $\mathcal{D}_{w,h}$  contain the matched errors at valid hour  $h$  during the four weeks preceding week  $w$ . For model  $m$ ,

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{w,h}|} \sum_{(f_m,y) \in \mathcal{D}_{w,h}} (f_m - y), \quad \tilde{f}_m(t) = f_m(t) - \bar{b}_{m,w,h(t)}. \quad (2)$$

The estimate is pooled across European stations and does not depend on initialisation or lead. For ensemble models, the same shift is applied to every member.

An additive correction is unsuitable for zero-bounded, right-skewed precipitation. Let  $\mathcal{P}_w$  contain forecasts and observations from the preceding four weeks, pooled across initialisations and leads. We apply

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left( \frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t). \quad (3)$$

This multiplicative correction adjusts systematic amount biases while preserving zero forecasts. Both procedures use only prior information and do not alter observation-defined regime membership or force the conditional error within each regime to zero.

## 2.5 EVALUATION PROTOCOL

We assess forecast accuracy relative to the high-resolution deterministic ECMWF IFS forecast. For a given variable, let  $c = (s, t_0, \tau)$  denote an evaluation cell at station  $s$ , initialisation time  $t_0$ , and lead time  $\tau$ , verified at  $t_0 + \tau$ . For model  $m$ , let  $\mathcal{C}_m$  denote the set of cells for which the bias-corrected forecast, the IFS forecast, and the verifying observation are all available. Each comparison therefore uses forecast–observation pairs matched in station, initialisation time, lead time, and valid time.

For the retained cells  $\mathcal{C}_m$ , MAE skill relative to IFS is

$$S_{m,\mathcal{C}_m} = 100 \left( 1 - \frac{\text{MAE}_{m,\mathcal{C}_m}}{\text{MAE}_{\text{IFS},\mathcal{C}_m}} \right). \quad (4)$$

Positive skill indicates lower MAE than IFS, whereas negative skill indicates higher MAE. Initialisation times are restricted to the 00, 06, 12, and 18 UTC cycles, and lower-cadence models contribute only at their native leads.

The evaluation covers thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. It is organised into two tracks. Track A covers 1 September 2025 to 30 June 2026. Wind and temperature are evaluated at 6-hourly leads from 6 to 48 h, whereas solar radiation and hourly precipitation are evaluated at all leads from 1 to 48 h. Track B covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU on a common hourly schedule at leads from 1 to 48 h; EPT-2.1 Helios is additionally included for solar radiation. Italy and Poland are excluded from precipitation because of missing observations.

Before scoring, forecasts are bilinearly interpolated to station locations (Molinaro et al., 2026). Precipitation is evaluated only at gauges within each model’s spatial domain, and regional comparisons use a common gauge sample. Ensemble models are evaluated through their means; the analysis therefore measures point-forecast accuracy rather than probabilistic calibration or ensemble spread.

None of the Jua models was trained or fine-tuned on the verification observations. Historical station errors enter only through the causal bias-correction procedure described above. Uncertainty in aggregated skill is quantified using a leave-one-month-out jackknife, with error bars representing one standard error.

### 3 RESULTS

#### 3.1 SKILL ACROSS THE OBSERVED DISTRIBUTION

Figure 1 shows that tail skill does not follow a uniform AI–physical ordering. EPT-2.1 Europa leads the wind comparison overall and remains competitive with ICON Global at gale force. EPT-2 HRRR and EPT-2.1 Europa perform most strongly for temperature, including in the heat regime. By contrast, AIFS and AIFS ENS remain below IFS in the wind and temperature tails despite the ensemble’s improved performance near the centre of the distribution. The largest heat deficit belongs to the physical GFS model.

Solar radiation follows a different pattern. EPT-2.1 Helios leads overall and under overcast and near-clear-sky conditions but falls below IFS under typical irradiance, where EPT-2 Reasoning performs most strongly. Its advantage is therefore concentrated in the tails rather than distributed uniformly across conditions.

For precipitation, the clearest gains occur in the middle of the wet-hour distribution (Figure 2). EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS from moderate through high rain, whereas the physical global models remain below the reference. These gains narrow at the heaviest intensities. Europa is the exception: its positive all-conditions score comes from dry hours, while it is worse than IFS in every wet-hour regime.

We additionally evaluate detection of heavy-rain events using the local wet-hour P95 threshold. For each forecast–observation pair, the observation and forecast are each labelled a heavy-rain event if their hourly accumulation exceeds the same local wet-hour P95 threshold. All evaluated precipitation models forecast 29–82% as many P95 exceedances as observed. Europa and ICON Global nevertheless achieve a slightly higher critical success index than IFS (Figure S7; Table S24).

#### 3.2 SHARED CONDITIONAL-ERROR PATTERN

Despite their differences in relative skill, all models exhibit the same conditional-error structure (Figure 3): low observations are overpredicted and high observations underpredicted. The variation across regimes is larger than the spread between models and is present for all four variables. A common tendency towards the centre of the observed distribution therefore coexists with model-specific differences in relative accuracy.

Because the regimes are defined using the verifying observations, this pattern does not by itself show that the models systematically suppress extreme values. The same relationship can arise from limited predictability, unresolved local variability, or differences between point observations and gridded forecasts.

#### 3.3 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

To complement the climatological-regime analysis, we evaluate five externally documented record-breaking temperature events and seven major European windstorms. Event windows and affected-country footprints were specified from external reports before scoring. A forecast sample is retained only when IFS and every Track A model included in the comparison share the same station, valid time, initialisation, and 6-hourly lead from 6 to 48 h. Each model therefore uses identical event windows and forecast–observation samples. Full event definitions, provenance, scoring details, and numerical results are reported in Appendix A.6 (Tables S25, and S26).

Europe shows positive skill in both event sets (Figure 4). During the temperature events, performance varies substantially across models, with GFS showing the largest deficit. During the windstorms, most models remain close to IFS, whereas AIFS, AIFS ENS, and GFS remain below the reference. These source-defined cases therefore support the model-specific pattern found in the climatological-regime analysis, rather than a uniform AI-specific loss of skill during extremes.

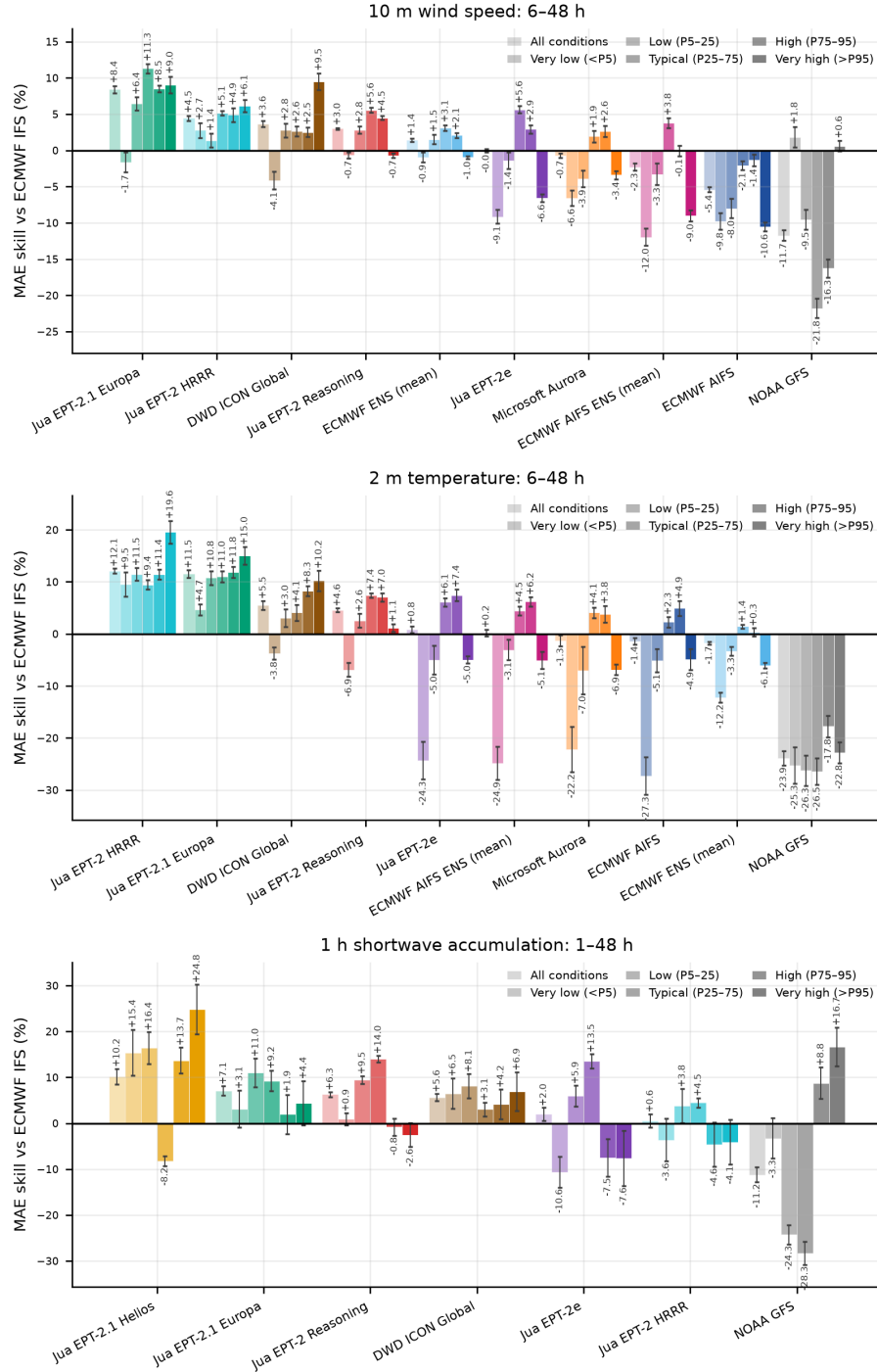


Figure 1: Track A MAE skill relative to ECMWF IFS by observed-value regime, with  $\pm 1$  jackknife SE. Wind and temperature use 6–48 h leads; solar radiation uses 1–48 h.

### 3.4 REGIONAL COMPARISON

Track B compares EPT-2.1 Europa and EPT-2 HRRR directly with ICON-EU over March–June 2026 (Figure 5). Europa retains the strongest overall wind estimate, while ICON-EU is competitive

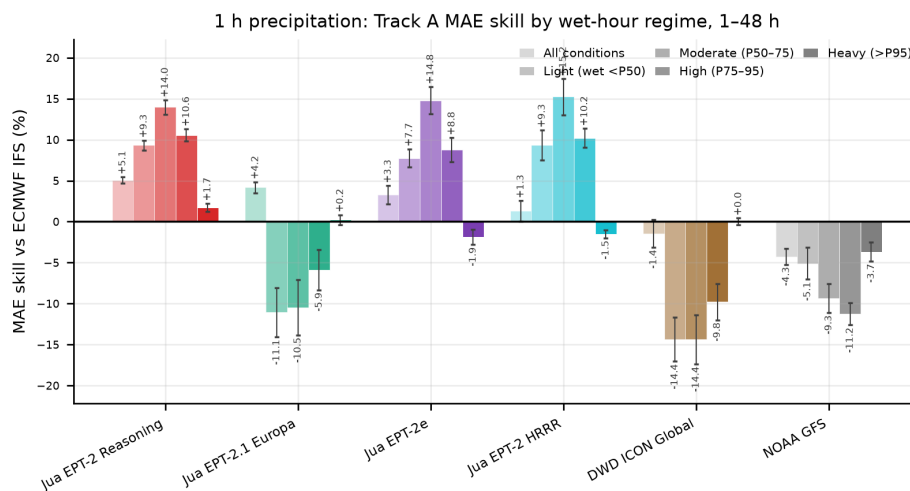


Figure 2: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime over 1–48 h, with  $\pm 1$  jackknife SE.

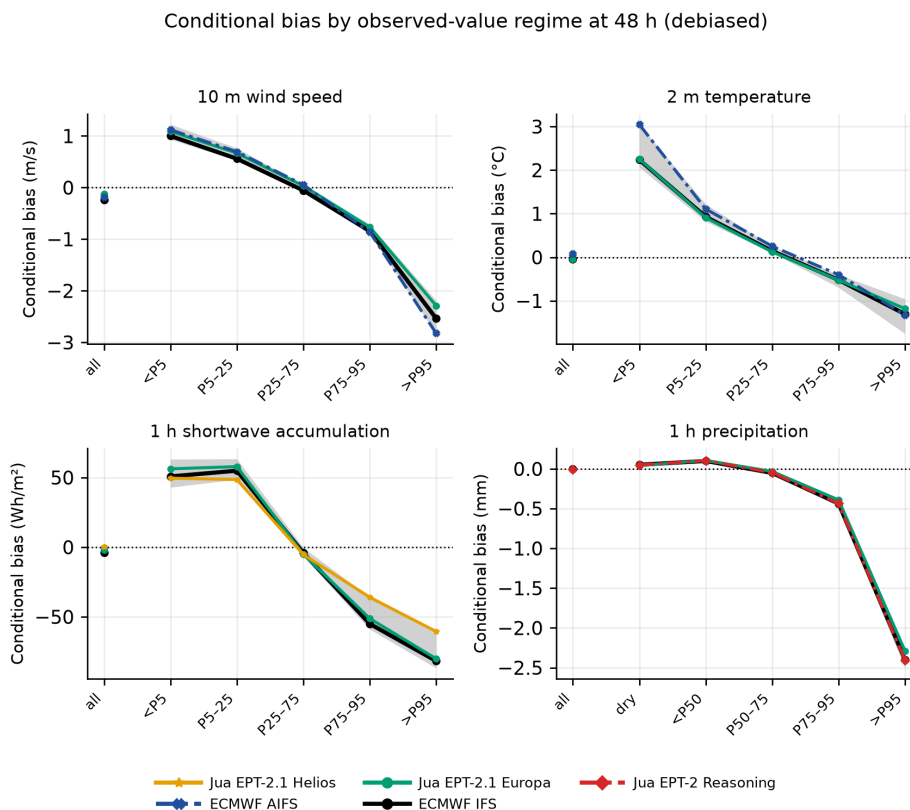


Figure 3: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models; selected models are highlighted.

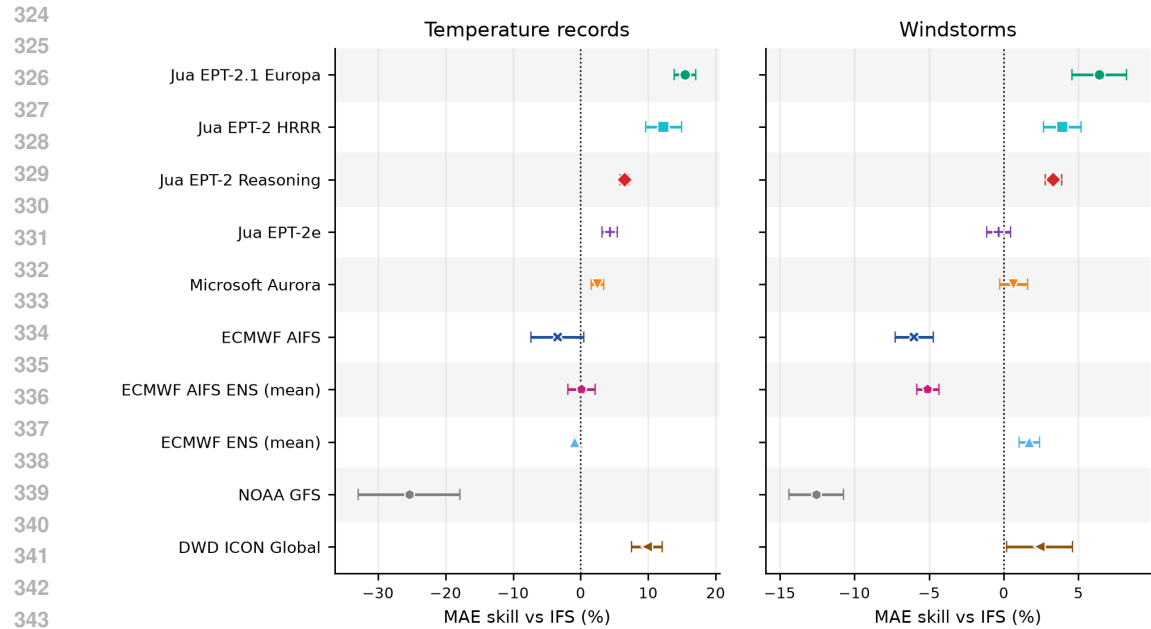


Figure 4: MAE skill relative to ECMWF IFS during five documented record-breaking temperature events (left;  $n = 231,065$ ) and seven major European windstorms (right;  $n = 123,847$ ), aggregated over common station/valid-time/lead samples from 6–48 h after bias correction. Error bars show one leave-one-event-out jackknife SE.

with the leading AI models in the wind and temperature tails. EPT-2.1 Helios has the clearest solar advantage.

The lead-time and 1–12 h analyses preserve the principal model ordering (Appendix A.2). In the matched regional comparison, ICON-EU remains competitive with the leading regional AI systems in the wind and temperature tails (Appendix A.3). Complete values are reported in Appendix A.4; the country profiles show that the main temperature and wind patterns are not driven by a single country (Appendix A.5).

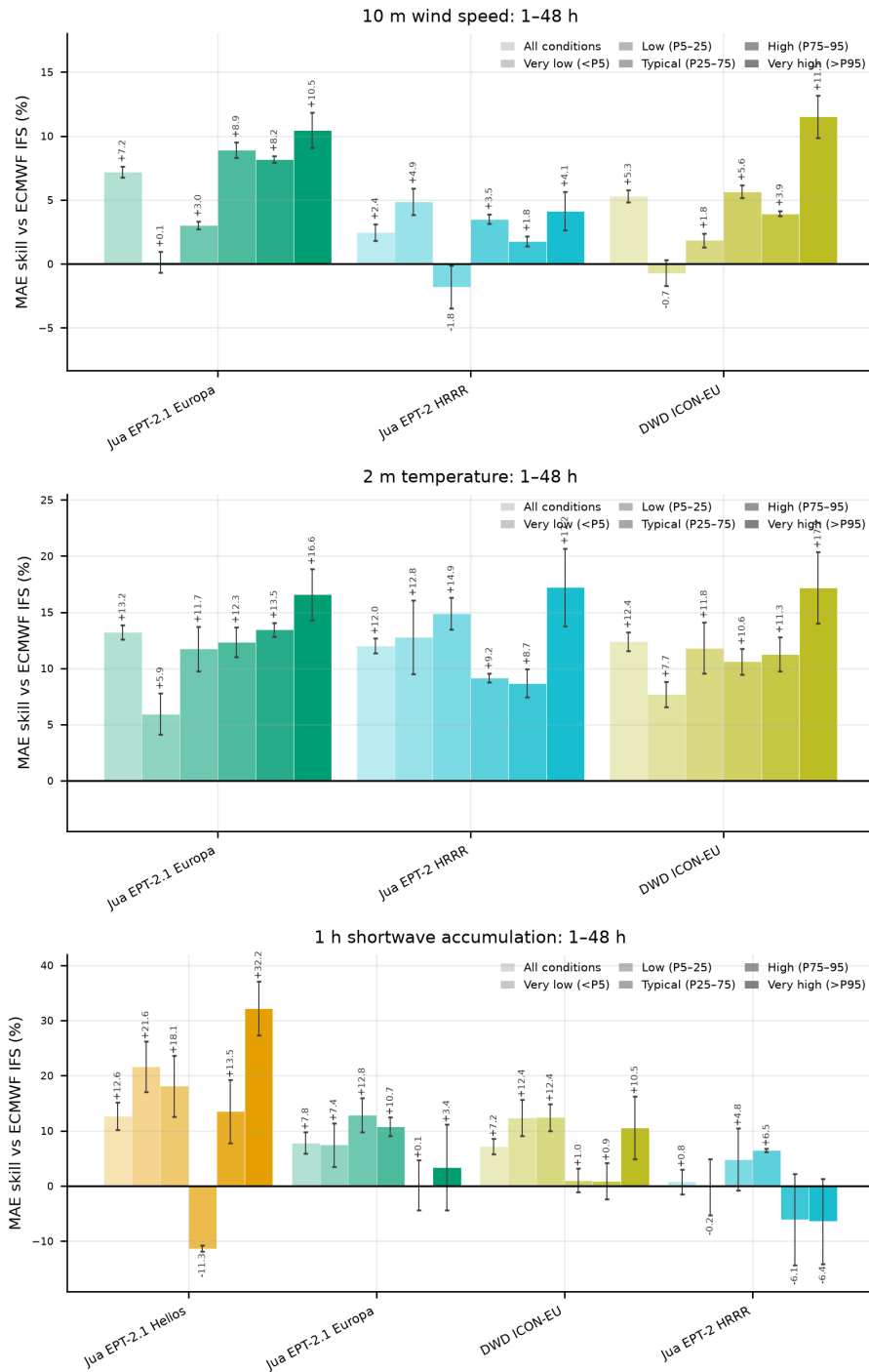


Figure 5: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with  $\pm 1$  jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

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## 4 DISCUSSION

The results distinguish between two questions: whether a model underestimates an extreme observation, and whether it is less accurate than IFS when that observation occurs. Across the evaluated models, low observations are overpredicted and high observations are underpredicted, although the magnitude of these errors varies. Several AI models are nevertheless closer to the observations than IFS in the heat, gale, solar, and precipitation regimes. Underestimation is therefore common to both AI and NWP models, whereas relative accuracy depends on the model, variable, and observed regime.

The same conditional-error pattern appears across deterministic, probabilistic, and generative AI models, as well as NWP models, with no consistent ordering between these classes. Instead, differences emerge at the level of individual systems: AIFS and AIFS ENS perform worse than IFS in the wind and temperature tails, whereas EPT-2 HRRR and EPT-2.1 Europa remain competitive at gale force and in heat. The NWP results likewise vary, with GFS showing the largest heat deficit while ICON-EU is competitive with the leading regional AI models in the matched Track B comparison. The source-defined temperature events and windstorms preserve this model-specific ordering on common event samples: the leading regional systems remain ahead of IFS, whereas the two AIFS variants and GFS lose skill in at least one event set.

Relative accuracy also depends on the observed regime and variable. For example, EPT-2.1 Helios is more accurate than IFS in both solar tails but less accurate under typical irradiance. Similarly, EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS for moderate to high precipitation, although their advantage narrows for the heaviest events.

Within these scope conditions, the results do not support a general weakness of AI weather forecasting at extremes.

### 4.1 LIMITATIONS

The comparisons identify differences in performance, but not their causes. The systems differ in architecture, training objective, resolution, training data, initialisation, cadence, and domain (Bonavita, 2024; Subich et al., 2025); consequently, their performance cannot be attributed independently to AI versus NWP, regression versus generative training, or global versus regional resolution. Moreover, evaluating ensembles through their means assesses point-forecast accuracy, rather than calibration, spread, or event probabilities.

### ETHICS STATEMENT

Several of the forecast systems evaluated in this study are developed by the authors' organisation. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the archived requests, aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

### REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates. The package is provided as anonymised supplementary material.

### AI USE STATEMENT

Generative AI tools were used to assist with software changes in the extraction, aggregation and figure code, to inspect generated results, and to draft and edit manuscript text. They were not used to generate or modify observations or forecast data. The authors checked AI-assisted code through

the pipeline’s executable self-tests and regenerated artifacts, verified the quoted numbers against the aggregate data, reviewed all manuscript text, and take full responsibility for the content.

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## A SUPPLEMENTARY MATERIAL

This supplement provides the model and regime definitions omitted from the main text, lead-time and short-range robustness checks for Track A, the matched regional comparison in Track B, complete numerical tables, and country-level regime profiles.

### A.1 EVALUATION SETUP

Surface wind and temperature observations were obtained through Synoptic Data from selected operational observing networks, including SYNOP, WIS2, ASOS/AWOS, and METAR. Solar-radiation observations were compiled from ARPALomb, BSRN, DMI, DWD, FMI, Frost, GeoSphere Austria, KNMI, Météo-France, MeteoSwiss, RMI, and SMHI.

Table S1 identifies the forecasting systems and their broad model classes. Tables S2 and S3 give the observed-value intervals and meteorological labels used throughout the evaluation.

**Table S1:** Forecast models included in the evaluation. ECMWF IFS is the reference model.

| Model                 | Model class           | Forecast type | Domain   |
|-----------------------|-----------------------|---------------|----------|
| ECMWF IFS (reference) | Physical              | Deterministic | Global   |
| ECMWF ENS             | Physical              | Ensemble      | Global   |
| NOAA GFS              | Physical              | Deterministic | Global   |
| DWD ICON Global       | Physical              | Deterministic | Global   |
| DWD ICON-EU           | Physical              | Deterministic | Regional |
| ECMWF AIFS            | Regression AI         | Deterministic | Global   |
| Microsoft Aurora      | Regression AI         | Deterministic | Global   |
| Jua EPT-2 Reasoning   | Regression AI         | Deterministic | Global   |
| Jua EPT-2e            | Regression AI         | Ensemble      | Global   |
| Jua EPT-2.1 Helios    | Regression AI (solar) | Deterministic | Regional |
| ECMWF AIFS ENS        | Probabilistic AI      | Ensemble      | Global   |
| Jua EPT-2 HRRR        | Generative AI         | Ensemble      | Regional |
| Jua EPT-2.1 Europa    | Generative AI         | Ensemble      | Regional |

**Table S2:** Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the percentiles refer to positive daytime GHI.

| Regime    | Definition         | Wind            | Temperature    | Solar            |
|-----------|--------------------|-----------------|----------------|------------------|
| All       | –                  | all conditions  | all conditions | all daytime      |
| Very low  | $y < P5$           | calm            | severe cold    | heavily overcast |
| Low       | $P5 \leq y < P25$  | light air       | cold           | overcast         |
| Typical   | $P25 \leq y < P75$ | moderate        | moderate       | mixed sky        |
| High      | $P75 \leq y < P95$ | fresh to strong | warm           | mostly sunny     |
| Very high | $y \geq P95$       | gale force      | heat           | near clear sky   |

**Table S3:** Observed-value regimes for hourly precipitation. Wet-hour percentiles are conditional on ERA5 accumulations of at least 0.1 mm.

| Regime   | Definition                    | Interpretation     |
|----------|-------------------------------|--------------------|
| All      | –                             | all hours          |
| Dry      | $y < 0.1$ mm                  | dry hour           |
| Light    | $0.1 \text{ mm} \leq y < P50$ | light rain         |
| Moderate | $P50 \leq y < P75$            | moderate rain      |
| High     | $P75 \leq y < P95$            | heavy rain         |
| Heavy    | $y \geq P95$                  | most intense hours |

## A.2 TRACK A LEAD-TIME, SHORT-RANGE, AND PRECIPITATION ROBUSTNESS

Figures S1, S2, S3, and S4 show skill as a function of forecast lead. Figure S5 repeats the regime comparison over 1–12 h, while Table S4 reports heat-regime skill by country. Figure S6, Table S5, and Figure S7 provide the short-range, full-horizon, and categorical precipitation checks; short-range precipitation values are in Table S12.

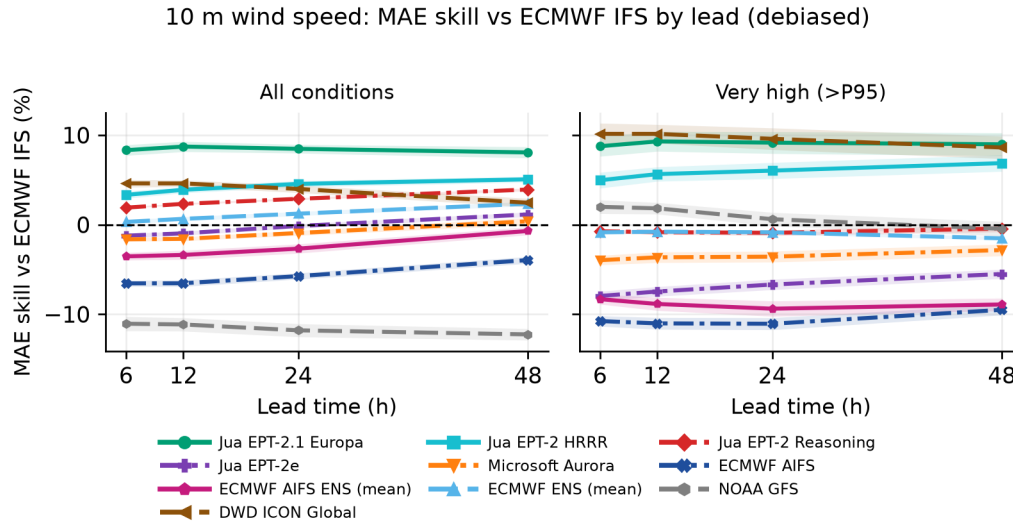


Figure S1: MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime,  $y \geq P95$ . Shading denotes  $\pm 1$  jackknife SE.

**Table S4:** Track A MAE skill (%) by country in the heat regime ( $y \geq P95$ ), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

| Model                    | DE    | FR    | GB    | ES    | IT    | PL    | NL    | BE    | CH    | NO    | AT    | CZ    | DK    |
|--------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Jua EPT-2.1 Europa       | +6.6  | +16.1 | -1.2  | +6.4  | +17.2 | +6.2  | -3.4  | -3.0  | +31.1 | +5.5  | +28.2 | +16.6 | -4.4  |
| Jua EPT-2 HRRR           | +10.5 | +21.5 | +4.0  | +10.5 | +27.6 | +5.6  | +1.8  | -0.6  | +29.4 | +14.2 | +16.6 | +17.7 | +12.2 |
| Jua EPT-2 Reasoning      | +6.6  | +2.6  | +4.3  | -2.6  | +0.1  | +3.9  | +3.8  | -1.0  | -1.3  | -5.2  | -3.1  | +7.5  | +2.7  |
| Jua EPT-2e               | +6.1  | -2.2  | -0.3  | -15.1 | -10.4 | -2.4  | +0.6  | -2.8  | -2.5  | -12.2 | -16.3 | -1.3  | +1.7  |
| Microsoft Aurora         | +2.1  | -4.5  | -7.5  | -18.7 | -7.8  | -0.9  | +0.7  | -11.8 | -3.3  | -18.0 | -15.4 | -7.0  | +1.8  |
| ECMWF AIFS               | +0.8  | -1.6  | -10.2 | -12.7 | -14.6 | +1.9  | -0.4  | +6.3  | +1.9  | -3.4  | -25.0 | -3.1  | +10.3 |
| ECMWF AIFS ENS (mean)    | +5.3  | -3.1  | -6.7  | -10.6 | -12.6 | -3.1  | -3.9  | +5.1  | +4.6  | -3.2  | -38.7 | +2.1  | +4.4  |
| ECMWF ENS (mean)         | -2.1  | -4.6  | -5.1  | -12.1 | -4.0  | -8.6  | +3.4  | -2.9  | -10.7 | -9.8  | -10.6 | -1.5  | -9.2  |
| NOAA GFS                 | -18.4 | -23.6 | -37.0 | -25.1 | -13.2 | -24.3 | -21.9 | -12.2 | -10.2 | -43.1 | -25.3 | -31.4 | -24.7 |
| DWD ICON Global          | +6.9  | +8.8  | -4.5  | +3.6  | +12.8 | +4.8  | -2.7  | +10.0 | +22.7 | +18.1 | +16.8 | +15.5 | -6.5  |
| DWD ICON-EU <sup>†</sup> | +13.1 | +18.4 | +1.3  | +7.4  | +20.1 | +9.1  | -2.6  | +14.1 | +34.6 | +6.6  | +28.1 | +17.8 | -0.7  |

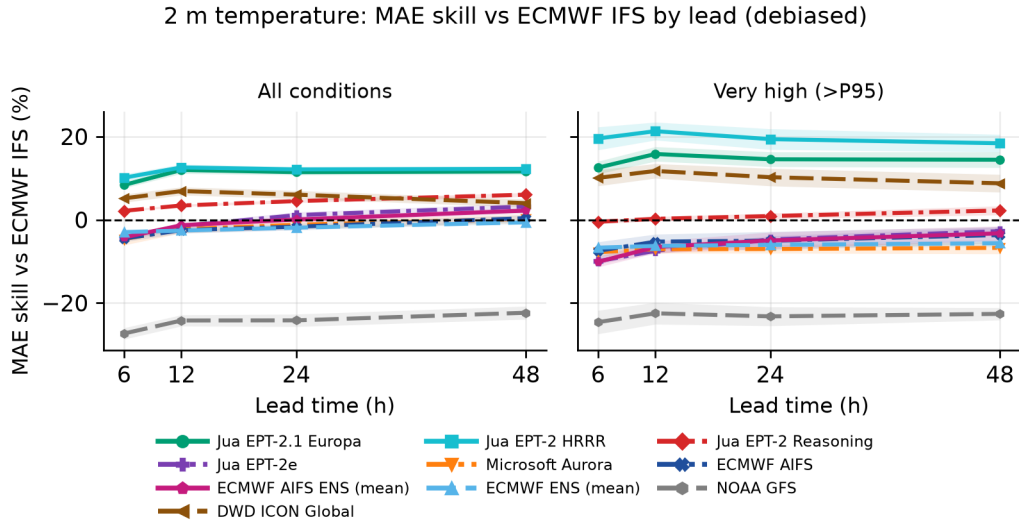


Figure S2: As Figure S1, for 2 m temperature. The right panel shows the heat regime,  $y \geq P95$ .

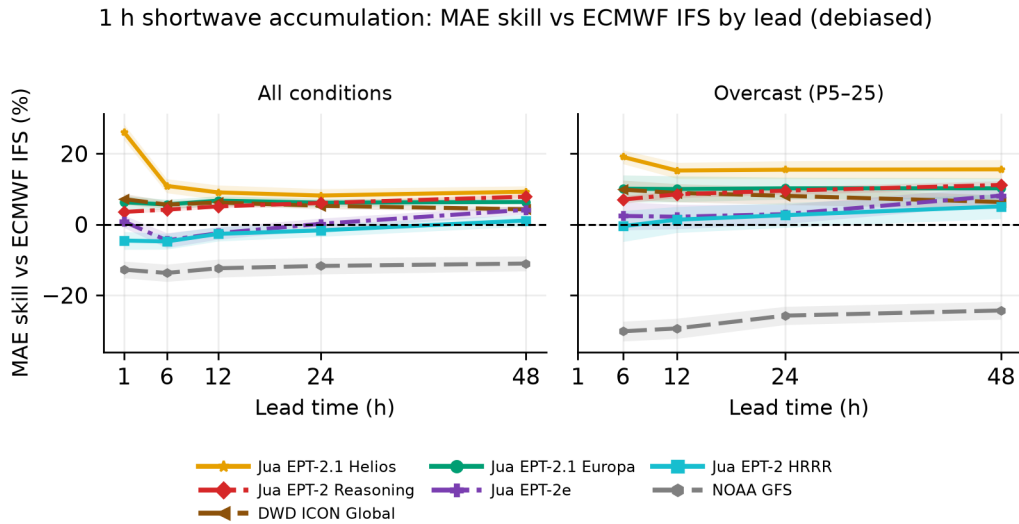


Figure S3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime,  $P5 \leq y < P25$ . AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes  $\pm 1$  jackknife SE.

**Table S5:** Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

| Model               | All                | Dry                 | <P50                | P50–75              | P75–95              | >P95               |
|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
| Jua EPT-2 Reasoning | +5.1 ( $\pm 0.4$ ) | +2.2 ( $\pm 1.3$ )  | +9.3 ( $\pm 0.6$ )  | +14.0 ( $\pm 0.9$ ) | +10.6 ( $\pm 0.8$ ) | +1.7 ( $\pm 0.5$ ) |
| Jua EPT-2 HRRR      | +1.3 ( $\pm 1.3$ ) | -6.1 ( $\pm 3.3$ )  | +9.3 ( $\pm 1.8$ )  | +15.2 ( $\pm 2.2$ ) | +10.2 ( $\pm 1.2$ ) | -1.5 ( $\pm 0.5$ ) |
| Jua EPT-2e          | +3.3 ( $\pm 1.1$ ) | +1.5 ( $\pm 4.8$ )  | +7.7 ( $\pm 1.1$ )  | +14.8 ( $\pm 1.7$ ) | +8.8 ( $\pm 1.5$ )  | -1.9 ( $\pm 0.9$ ) |
| Jua EPT-2.1 Europa  | +4.2 ( $\pm 0.7$ ) | +21.0 ( $\pm 2.7$ ) | -11.1 ( $\pm 3.0$ ) | -10.5 ( $\pm 3.4$ ) | -5.9 ( $\pm 2.5$ )  | +0.2 ( $\pm 0.6$ ) |
| NOAA GFS            | -4.3 ( $\pm 1.0$ ) | -0.2 ( $\pm 1.8$ )  | -5.1 ( $\pm 2.0$ )  | -9.3 ( $\pm 1.8$ )  | -11.2 ( $\pm 1.4$ ) | -3.7 ( $\pm 1.2$ ) |
| DWD ICON Global     | -1.4 ( $\pm 1.7$ ) | +8.3 ( $\pm 2.7$ )  | -14.4 ( $\pm 2.7$ ) | -14.4 ( $\pm 3.0$ ) | -9.8 ( $\pm 2.2$ )  | +0.0 ( $\pm 0.4$ ) |

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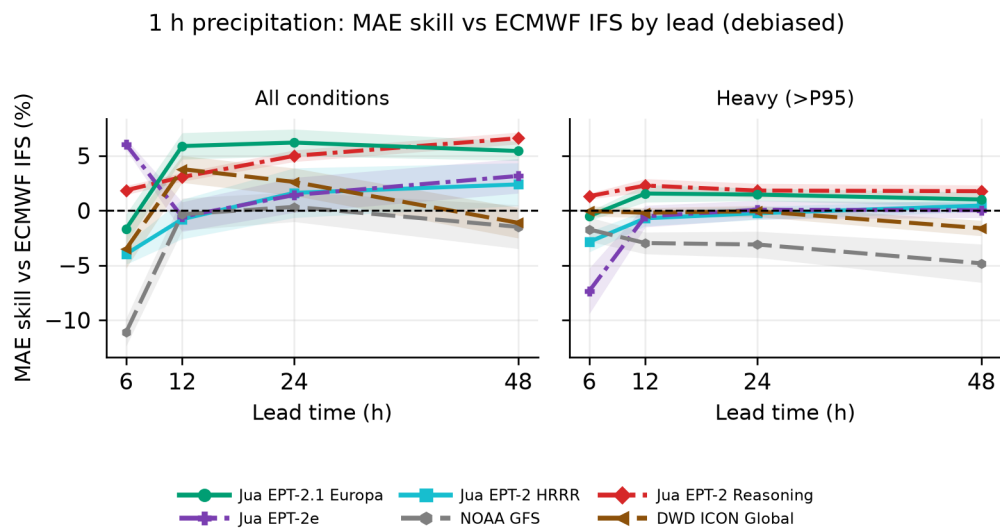


Figure S4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime,  $y \geq P95$ . Shading denotes  $\pm 1$  jackknife SE.

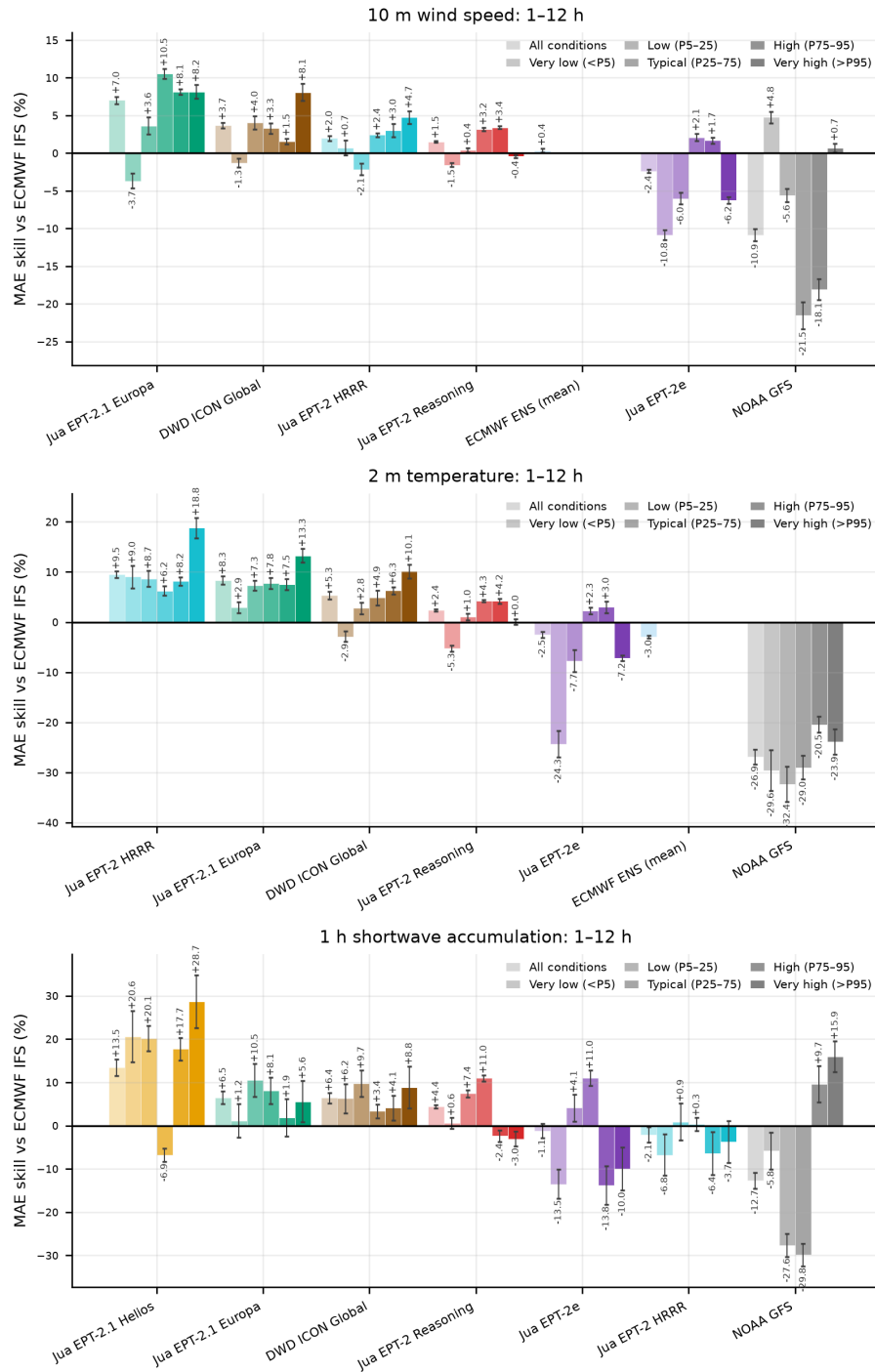


Figure S5: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with  $\pm 1$  jack-knife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

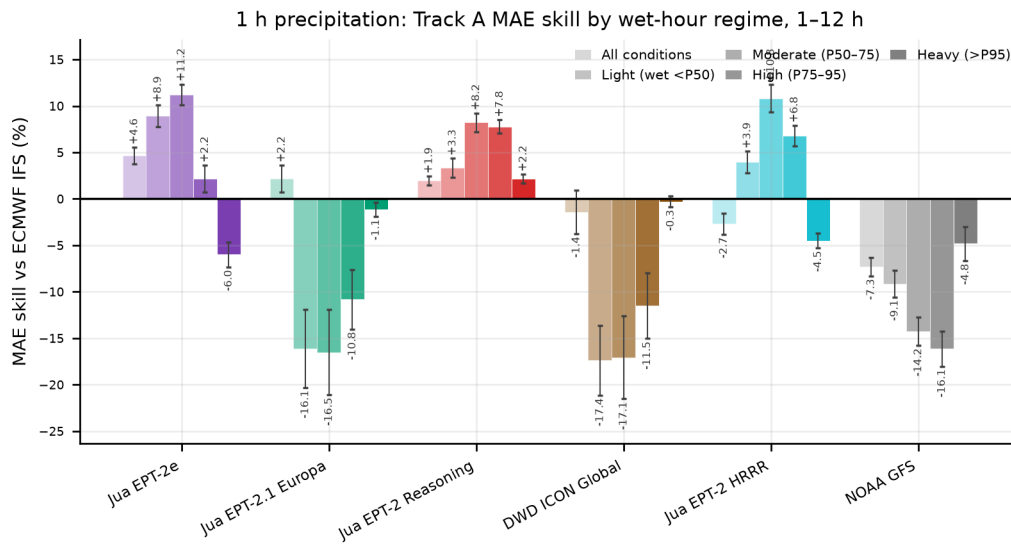


Figure S6: As Figure 2, restricted to the 1–12 h forecast range.

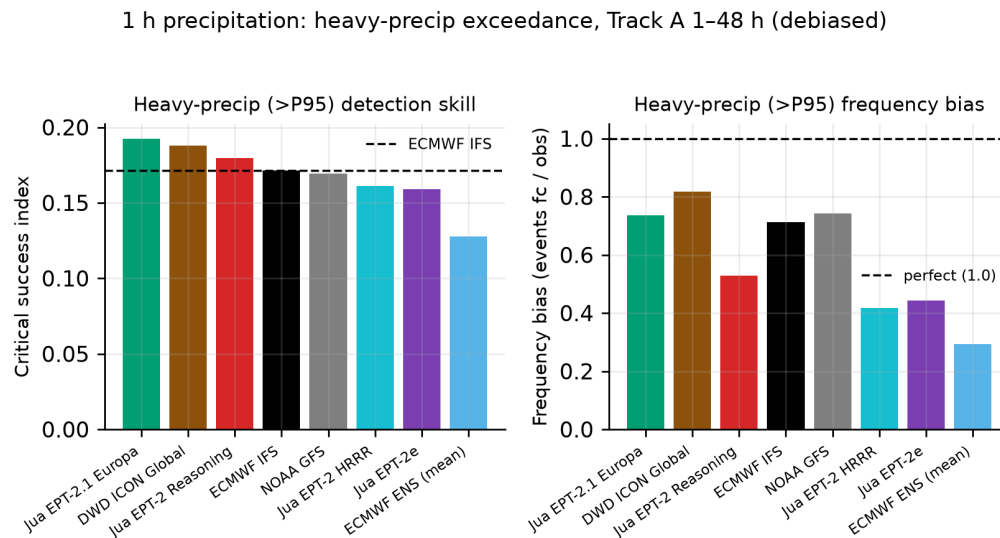


Figure S7: Detection of heavy precipitation ( $y \geq P95$ ) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

### A.3 MATCHED REGIONAL COMPARISON

Track B robustness beyond the full-horizon comparison in the main text is shown in Figure S8 for 1–12 h and in Figure S9 as a function of lead. Figures S10 and S11 give the corresponding precipitation results.

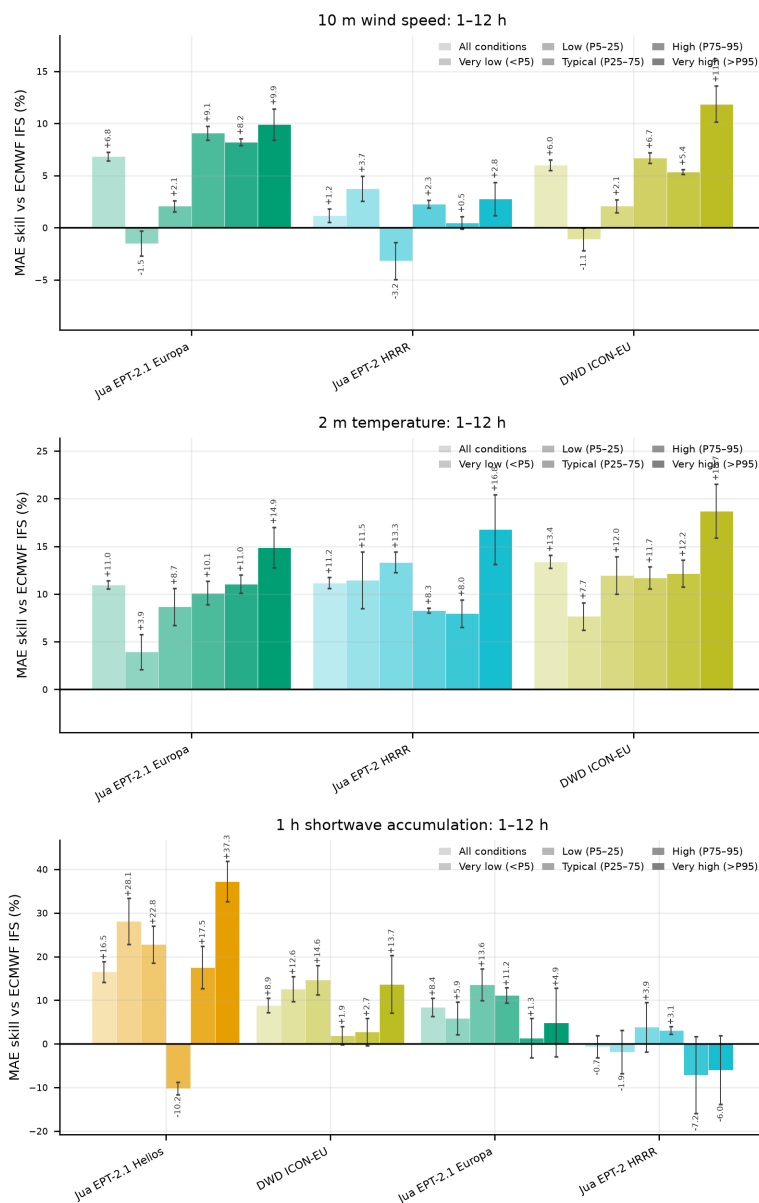


Figure S8: As Figure 5, restricted to the 1–12 h forecast range.

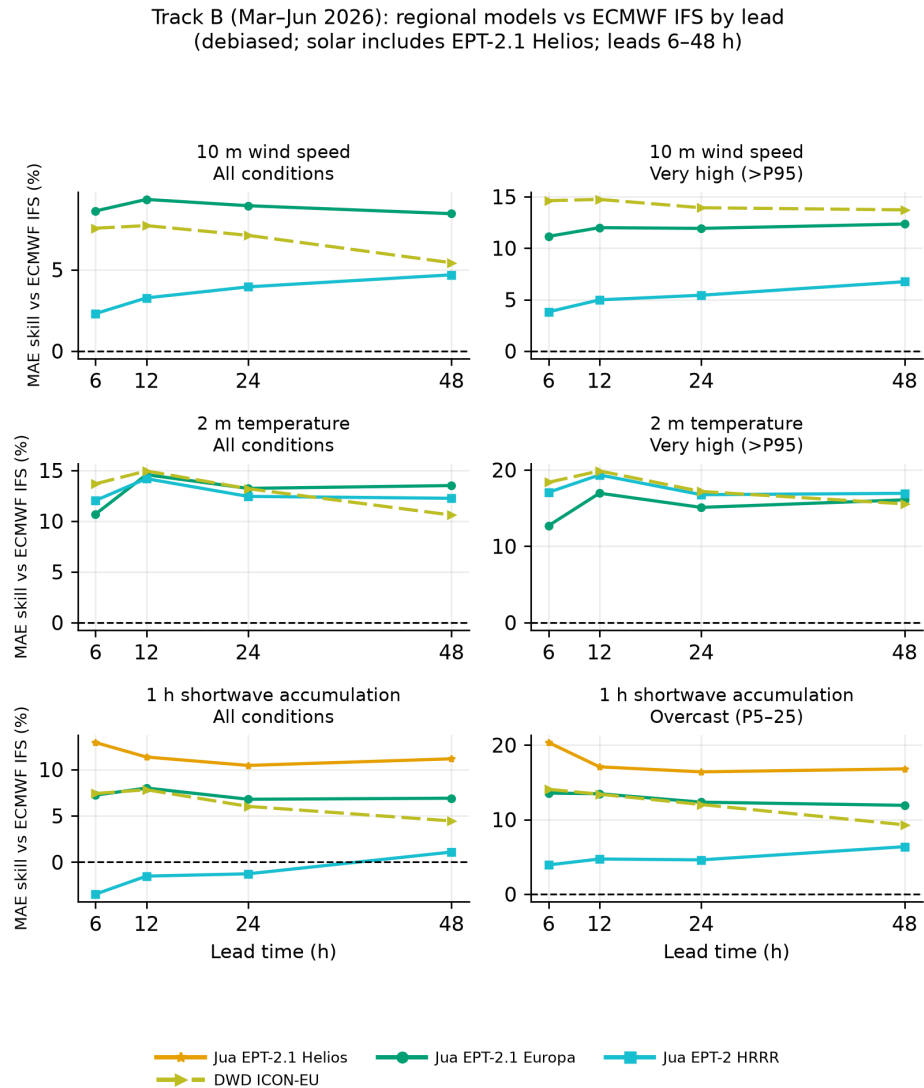


Figure S9: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes  $\pm 1$  jackknife SE.

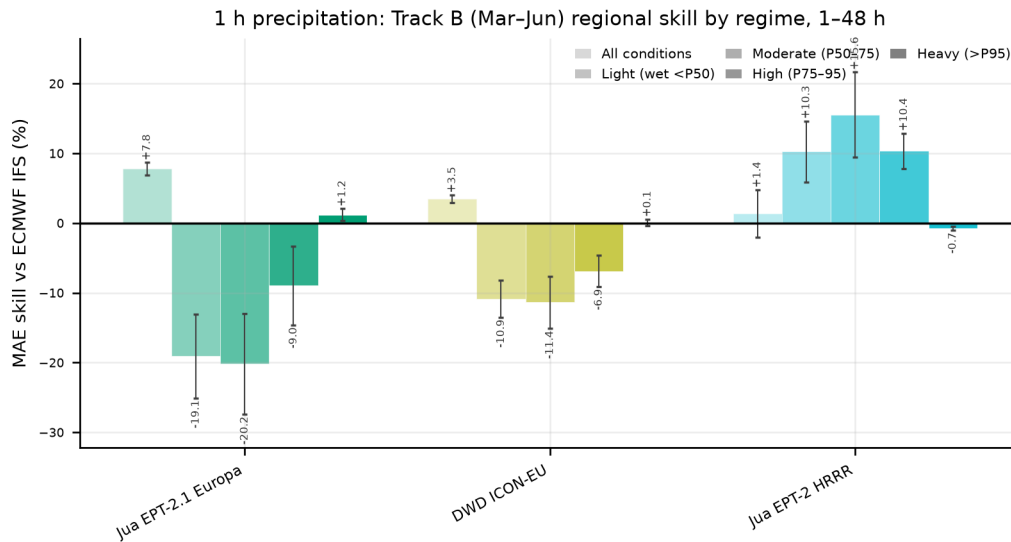


Figure S10: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with  $\pm 1$  jackknife SE.

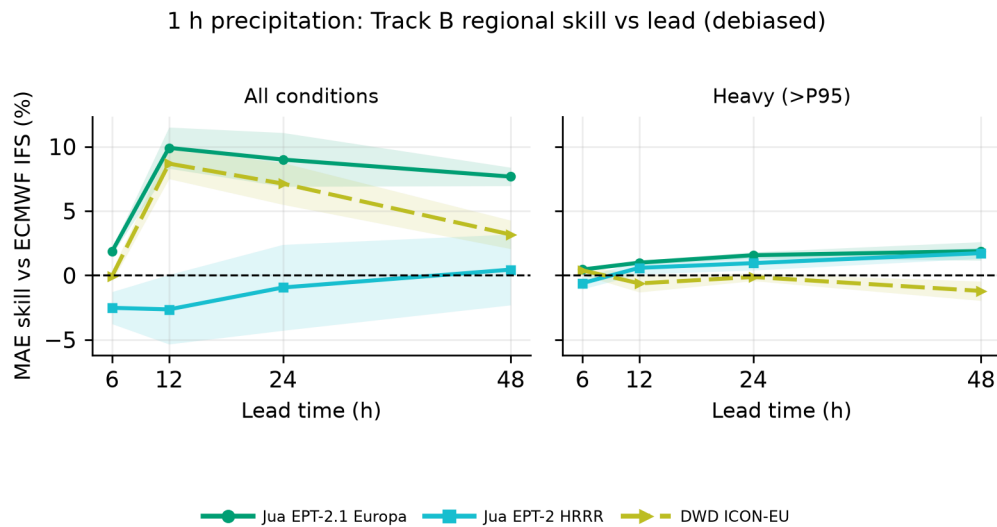


Figure S11: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime,  $y \geq P95$ . Shading denotes  $\pm 1$  jackknife SE.

#### A.4 COMPLETE NUMERICAL RESULTS

Track A full-horizon regime values are in Tables S6, S7, and S8; 1–12 h values are in Tables S9, S10, S11, and S12; per-lead values are in Tables S13, S14, and S15. Track B regime values are in Tables S16, S17, S18, and S23; its per-lead values are in Tables S19, S20, S21, and S22. Table S24 gives the heavy-precipitation detection metrics.

**Table S6:** Track A MAE skill (%), 6–48 h, by regime – Figure 1, wind. Parentheses: jackknife SE.

| Model                 | All         | <P5         | P5–25      | P25–75      | P75–95      | >P95        |
|-----------------------|-------------|-------------|------------|-------------|-------------|-------------|
| Jua EPT-2.1 Europa    | +8.4 (0.5)  | -1.7 (1.4)  | +6.4 (0.9) | +11.3 (0.7) | +8.5 (0.4)  | +9.0 (1.2)  |
| Jua EPT-2 HRRR        | +4.5 (0.3)  | +2.7 (1.0)  | +1.4 (0.9) | +5.1 (0.3)  | +4.9 (1.0)  | +6.1 (0.9)  |
| Jua EPT-2 Reasoning   | +3.0 (0.1)  | -0.7 (0.4)  | +2.8 (0.5) | +5.6 (0.3)  | +4.5 (0.3)  | -0.7 (0.3)  |
| Jua EPT-2e            | -0.0 (0.2)  | -9.1 (1.0)  | -1.4 (1.1) | +5.6 (0.5)  | +2.9 (0.6)  | -6.6 (0.5)  |
| Microsoft Aurora      | -0.7 (0.3)  | -6.6 (1.1)  | -3.9 (1.1) | +1.9 (0.8)  | +2.6 (0.8)  | -3.4 (0.6)  |
| ECMWF AIFS            | -5.4 (0.4)  | -9.8 (1.2)  | -8.0 (1.3) | -2.1 (0.6)  | -1.4 (0.8)  | -10.6 (0.6) |
| ECMWF AIFS ENS (mean) | -2.3 (0.5)  | -12.0 (1.2) | -3.3 (1.5) | +3.8 (0.7)  | -0.1 (0.7)  | -9.0 (0.7)  |
| ECMWF ENS (mean)      | +1.4 (0.2)  | -0.9 (0.7)  | +1.5 (0.6) | +3.1 (0.4)  | +2.1 (0.4)  | -1.0 (0.2)  |
| NOAA GFS              | -11.7 (0.7) | +1.8 (1.4)  | -9.5 (1.4) | -21.8 (1.4) | -16.3 (1.3) | +0.6 (0.7)  |
| DWD ICON Global       | +3.6 (0.4)  | -4.1 (1.2)  | +2.8 (1.0) | +2.6 (0.7)  | +2.5 (0.7)  | +9.5 (1.1)  |

**Table S7:** Track A MAE skill (%), 6–48 h – Figure 1, temperature.

| Model                 | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Europa    | +11.5 (0.7) | +4.7 (1.0)  | +10.8 (1.3) | +11.0 (1.0) | +11.8 (1.1) | +15.0 (1.7) |
| Jua EPT-2 HRRR        | +12.1 (0.4) | +9.5 (2.4)  | +11.5 (1.2) | +9.4 (0.9)  | +11.4 (1.0) | +19.6 (2.2) |
| Jua EPT-2 Reasoning   | +4.6 (0.4)  | -6.9 (1.3)  | +2.6 (1.3)  | +7.4 (0.4)  | +7.0 (0.8)  | +1.1 (0.7)  |
| Jua EPT-2e            | +0.8 (0.7)  | -24.3 (3.6) | -5.0 (2.8)  | +6.1 (0.8)  | +7.4 (1.1)  | -5.0 (0.7)  |
| Microsoft Aurora      | -1.3 (1.1)  | -22.2 (4.4) | -7.0 (4.6)  | +4.1 (1.0)  | +3.8 (1.6)  | -6.9 (1.0)  |
| ECMWF AIFS            | -1.4 (0.6)  | -27.3 (3.6) | -5.1 (2.3)  | +2.3 (1.0)  | +4.9 (1.4)  | -4.9 (2.0)  |
| ECMWF AIFS ENS (mean) | +0.2 (0.6)  | -24.9 (3.2) | -3.1 (2.0)  | +4.5 (0.9)  | +6.2 (0.9)  | -5.1 (1.7)  |
| ECMWF ENS (mean)      | -1.7 (0.3)  | -12.2 (1.0) | -3.3 (0.9)  | +1.4 (0.4)  | +0.3 (0.8)  | -6.1 (0.5)  |
| NOAA GFS              | -23.9 (1.4) | -25.3 (3.5) | -26.3 (2.9) | -26.5 (2.5) | -17.8 (2.1) | -22.8 (2.0) |
| DWD ICON Global       | +5.5 (0.9)  | -3.8 (1.2)  | +3.0 (1.7)  | +4.1 (1.5)  | +8.3 (0.9)  | +10.2 (2.0) |

**Table S8:** Track A MAE skill (%), 1–48 h hourly – Figure 1, solar. Parentheses: jackknife SE.

| Model               | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|---------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Helios  | +10.2 (1.7) | +15.4 (5.0) | +16.4 (3.4) | -8.2 (1.1)  | +13.7 (2.9) | +24.8 (5.4) |
| Jua EPT-2.1 Europa  | +7.1 (1.0)  | +3.1 (4.0)  | +11.0 (3.1) | +9.2 (2.3)  | +1.9 (4.3)  | +4.4 (4.8)  |
| Jua EPT-2 HRRR      | +0.6 (1.4)  | -3.6 (4.6)  | +3.8 (3.7)  | +4.5 (1.0)  | -4.6 (4.8)  | -4.1 (4.9)  |
| Jua EPT-2 Reasoning | +6.3 (0.5)  | +0.9 (1.3)  | +9.5 (0.8)  | +14.0 (0.7) | -0.8 (1.9)  | -2.6 (2.6)  |
| Jua EPT-2e          | +2.0 (1.4)  | -10.6 (3.3) | +5.9 (2.3)  | +13.5 (1.6) | -7.5 (4.1)  | -7.6 (6.0)  |
| NOAA GFS            | -11.2 (1.6) | -3.3 (4.4)  | -24.3 (2.1) | -28.3 (2.5) | +8.8 (3.4)  | +16.7 (4.2) |
| DWD ICON Global     | +5.6 (0.8)  | +6.5 (3.3)  | +8.1 (2.6)  | +3.1 (1.5)  | +4.2 (3.3)  | +6.9 (4.2)  |

**Table S9:** Track A MAE skill (%), 1–12 h – Figure S5, wind.

| Model               | All         | <P5         | P5–25      | P25–75      | P75–95      | >P95       |
|---------------------|-------------|-------------|------------|-------------|-------------|------------|
| Jua EPT-2.1 Europa  | +7.0 (0.5)  | -3.7 (1.0)  | +3.6 (1.1) | +10.5 (0.7) | +8.1 (0.3)  | +8.2 (0.9) |
| Jua EPT-2 HRRR      | +2.0 (0.3)  | +0.7 (1.0)  | -2.1 (0.8) | +2.4 (0.2)  | +3.0 (0.9)  | +4.7 (0.9) |
| Jua EPT-2 Reasoning | +1.5 (0.1)  | -1.5 (0.2)  | +0.4 (0.3) | +3.2 (0.2)  | +3.4 (0.2)  | -0.4 (0.2) |
| Jua EPT-2e          | -2.4 (0.2)  | -10.8 (0.7) | -6.0 (0.7) | +2.1 (0.5)  | +1.7 (0.4)  | -6.2 (0.4) |
| ECMWF ENS (mean)    | +0.4 (0.3)  | –           | –          | –           | –           | –          |
| NOAA GFS            | -10.9 (0.8) | +4.8 (0.8)  | -5.6 (0.9) | -21.5 (1.8) | -18.1 (1.4) | +0.7 (0.6) |
| DWD ICON Global     | +3.7 (0.3)  | -1.3 (0.6)  | +4.0 (0.9) | +3.3 (0.7)  | +1.5 (0.4)  | +8.1 (1.1) |

**Table S10:** Track A MAE skill (%), 1–12 h – Figure S5, temperature.

| Model               | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|---------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Europa  | +8.3 (0.8)  | +2.9 (1.1)  | +7.3 (1.0)  | +7.8 (1.1)  | +7.5 (1.1)  | +13.3 (1.4) |
| Jua EPT-2 HRRR      | +9.5 (0.6)  | +9.0 (2.3)  | +8.7 (1.6)  | +6.2 (1.0)  | +8.2 (0.8)  | +18.8 (2.0) |
| Jua EPT-2 Reasoning | +2.4 (0.2)  | -5.3 (0.6)  | +1.0 (0.6)  | +4.3 (0.2)  | +4.2 (0.5)  | +0.0 (0.5)  |
| Jua EPT-2e          | -2.5 (0.6)  | -24.3 (2.7) | -7.7 (2.2)  | +2.3 (0.7)  | +3.0 (1.1)  | -7.2 (0.5)  |
| ECMWF ENS (mean)    | -3.0 (0.2)  | –           | –           | –           | –           | –           |
| NOAA GFS            | -26.9 (1.5) | -29.6 (4.0) | -32.4 (3.5) | -29.0 (2.4) | -20.5 (1.6) | -23.9 (2.5) |
| DWD ICON Global     | +5.3 (0.8)  | -2.9 (1.0)  | +2.8 (1.2)  | +4.9 (1.5)  | +6.3 (0.7)  | +10.1 (1.4) |

**Table S11:** Track A MAE skill (%), 1–12 h – Figure S5, solar.

| Model               | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|---------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Helios  | +13.5 (1.9) | +20.6 (5.9) | +20.1 (2.9) | -6.9 (1.5)  | +17.7 (2.6) | +28.7 (6.1) |
| Jua EPT-2.1 Europa  | +6.5 (1.4)  | +1.2 (3.9)  | +10.5 (3.8) | +8.1 (3.0)  | +1.9 (4.3)  | +5.6 (4.8)  |
| Jua EPT-2 HRRR      | -2.1 (1.8)  | -6.8 (4.7)  | +0.9 (4.3)  | +0.3 (1.5)  | -6.4 (5.0)  | -3.7 (4.8)  |
| Jua EPT-2 Reasoning | +4.4 (0.4)  | +0.6 (1.3)  | +7.4 (0.8)  | +11.0 (0.7) | -2.4 (1.3)  | -3.0 (1.7)  |
| Jua EPT-2e          | -1.1 (1.7)  | -13.5 (3.4) | +4.1 (3.1)  | +11.0 (1.7) | -13.8 (4.5) | -10.0 (5.0) |
| NOAA GFS            | -12.7 (1.9) | -5.8 (4.2)  | -27.6 (2.7) | -29.8 (2.6) | +9.7 (4.2)  | +15.9 (3.6) |
| DWD ICON Global     | +6.4 (1.2)  | +6.2 (3.4)  | +9.7 (3.0)  | +3.4 (1.6)  | +4.1 (2.9)  | +8.8 (4.8)  |

**Table S12:** Track A precipitation MAE skill (%), 1–12 h (Figure S6). Parentheses: jackknife SE.

| Model               | All                | Dry                 | <P50                | P50–75              | P75–95              | >P95               |
|---------------------|--------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
| Jua EPT-2 Reasoning | +1.9 ( $\pm 0.5$ ) | -2.9 ( $\pm 1.0$ )  | +3.3 ( $\pm 1.0$ )  | +8.2 ( $\pm 1.0$ )  | +7.8 ( $\pm 0.7$ )  | +2.2 ( $\pm 0.5$ ) |
| Jua EPT-2 HRRR      | -2.7 ( $\pm 1.1$ ) | -10.9 ( $\pm 3.8$ ) | +3.9 ( $\pm 1.2$ )  | +10.8 ( $\pm 1.5$ ) | +6.8 ( $\pm 1.1$ )  | -4.5 ( $\pm 0.8$ ) |
| Jua EPT-2e          | +4.6 ( $\pm 0.9$ ) | +12.4 ( $\pm 3.4$ ) | +8.9 ( $\pm 1.1$ )  | +11.2 ( $\pm 1.1$ ) | +2.2 ( $\pm 1.5$ )  | -6.0 ( $\pm 1.3$ ) |
| Jua EPT-2.1 Europa  | +2.2 ( $\pm 1.5$ ) | +22.0 ( $\pm 3.2$ ) | -16.1 ( $\pm 4.2$ ) | -16.5 ( $\pm 4.6$ ) | -10.8 ( $\pm 3.2$ ) | -1.1 ( $\pm 0.7$ ) |
| NOAA GFS            | -7.3 ( $\pm 1.0$ ) | -3.4 ( $\pm 3.6$ )  | -9.1 ( $\pm 1.4$ )  | -14.2 ( $\pm 1.5$ ) | -16.1 ( $\pm 1.9$ ) | -4.8 ( $\pm 1.8$ ) |
| DWD ICON Global     | -1.4 ( $\pm 2.3$ ) | +11.4 ( $\pm 3.3$ ) | -17.4 ( $\pm 3.8$ ) | -17.1 ( $\pm 4.5$ ) | -11.5 ( $\pm 3.5$ ) | -0.3 ( $\pm 0.6$ ) |

**Table S13:** Track A MAE skill (%) by lead – Figure S1, wind. Point estimates (no SE in per-lead tables).

| Model           | 6 h   |       | 12 h  |       | 24 h  |       | 48 h  |      |
|-----------------|-------|-------|-------|-------|-------|-------|-------|------|
|                 | All   | >P95  | All   | >P95  | All   | >P95  | All   | >P95 |
| EPT-2.1 Europa  | +8.3  | +8.8  | +8.7  | +9.3  | +8.5  | +9.2  | +8.1  | +9.0 |
| EPT-2 HRRR      | +3.3  | +5.0  | +3.9  | +5.6  | +4.6  | +6.0  | +5.1  | +6.9 |
| EPT-2 Reasoning | +1.9  | -0.7  | +2.3  | -0.8  | +2.9  | -0.9  | +3.9  | -0.4 |
| EPT-2e          | -1.2  | -8.0  | -1.0  | -7.5  | -0.1  | -6.7  | +1.1  | -5.5 |
| Aurora          | -1.6  | -4.0  | -1.6  | -3.6  | -0.9  | -3.6  | +0.4  | -2.8 |
| AIFS            | -6.6  | -10.8 | -6.5  | -11.0 | -5.7  | -11.1 | -4.0  | -9.5 |
| AIFS ENS        | -3.5  | -8.3  | -3.4  | -8.9  | -2.7  | -9.4  | -0.7  | -8.9 |
| ENS             | +0.3  | -0.9  | +0.7  | -0.8  | +1.2  | -0.9  | +2.3  | -1.5 |
| GFS             | -11.1 | +2.0  | -11.1 | +1.8  | -11.8 | +0.6  | -12.3 | -0.5 |
| ICON Global     | +4.6  | +10.1 | +4.6  | +10.1 | +4.0  | +9.6  | +2.4  | +8.6 |

**Table S14:** Track A MAE skill (%) by lead – Figure S2, temperature. Point estimates (no SE in per-lead tables).

| Model           | 6 h   |       | 12 h  |       | 24 h  |       | 48 h  |       |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|
|                 | All   | >P95  | All   | >P95  | All   | >P95  | All   | >P95  |
| EPT-2.1 Europa  | +8.4  | +12.5 | +12.0 | +15.8 | +11.5 | +14.5 | +11.6 | +14.4 |
| EPT-2 HRRR      | +10.1 | +19.5 | +12.6 | +21.3 | +12.1 | +19.4 | +12.2 | +18.4 |
| EPT-2 Reasoning | +2.2  | -0.5  | +3.4  | +0.2  | +4.5  | +0.9  | +6.0  | +2.2  |
| EPT-2e          | -4.2  | -10.0 | -1.5  | -7.5  | +1.1  | -4.7  | +3.2  | -2.7  |
| Aurora          | -4.6  | -7.8  | -2.3  | -7.1  | -1.2  | -7.0  | +0.2  | -6.7  |
| AIFS            | -4.5  | -7.5  | -2.5  | -5.2  | -1.7  | -5.0  | +0.4  | -3.7  |
| AIFS ENS        | -4.3  | -10.1 | -1.3  | -6.6  | +0.1  | -5.0  | +2.2  | -3.3  |
| ENS             | -3.0  | -6.7  | -2.6  | -6.3  | -1.9  | -6.0  | -0.6  | -5.6  |
| GFS             | -27.3 | -24.6 | -24.2 | -22.5 | -24.2 | -23.2 | -22.3 | -22.6 |
| ICON Global     | +5.1  | +10.1 | +6.9  | +11.7 | +6.0  | +10.2 | +4.0  | +8.8  |

**Table S15:** Track A MAE skill (%) by lead – Figure S3, solar (Ovc. = overcast, P5–P25). Point estimates; SEs in regime tables.

| Model           | 1 h   |      | 6 h   |       | 12 h  |       | 24 h  |       | 48 h  |       |
|-----------------|-------|------|-------|-------|-------|-------|-------|-------|-------|-------|
|                 | All   | Ovc. | All   | Ovc.  | All   | Ovc.  | All   | Ovc.  | All   | Ovc.  |
| EPT-2.1 Helios  | +26.1 | –    | +10.9 | +19.0 | +9.0  | +15.2 | +8.1  | +15.4 | +9.2  | +15.5 |
| EPT-2.1 Europa  | +6.2  | –    | +5.6  | +10.1 | +6.7  | +10.0 | +6.1  | +10.2 | +6.3  | +10.2 |
| ICON Global     | +7.1  | –    | +5.6  | +9.8  | +6.2  | +8.9  | +5.2  | +8.1  | +4.3  | +6.3  |
| EPT-2 Reasoning | +3.5  | –    | +4.2  | +7.0  | +5.1  | +8.5  | +6.0  | +9.5  | +7.8  | +11.1 |
| EPT-2e          | +0.8  | –    | -4.5  | +2.4  | -2.5  | +2.2  | +0.2  | +2.9  | +4.1  | +8.1  |
| EPT-2 HRRR      | -4.6  | –    | -4.8  | -0.5  | -2.7  | +1.3  | -1.7  | +2.6  | +1.1  | +5.1  |
| GFS             | -12.8 | –    | -13.7 | -30.2 | -12.4 | -29.4 | -11.7 | -25.8 | -11.1 | -24.3 |

**Table S16:** Track B wind MAE skill (%), 1–48 h, by regime (Figure 5).

| Model              | All        | <P5        | P5–25      | P25–75     | P75–95     | >P95        |
|--------------------|------------|------------|------------|------------|------------|-------------|
| Jua EPT-2.1 Europa | +7.2 (0.4) | +0.1 (0.8) | +3.0 (0.3) | +8.9 (0.6) | +8.2 (0.3) | +10.5 (1.4) |
| Jua EPT-2 HRRR     | +2.4 (0.6) | +4.9 (1.0) | -1.8 (1.7) | +3.5 (0.4) | +1.8 (0.4) | +4.1 (1.5)  |
| DWD ICON-EU        | +5.3 (0.5) | -0.7 (1.0) | +1.8 (0.6) | +5.6 (0.5) | +3.9 (0.2) | +11.5 (1.6) |

**Table S17:** Track B MAE skill (%), 1–48 h – Figure 5, temperature.

| Model              | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Europa | +13.2 (0.6) | +5.9 (1.8)  | +11.7 (2.0) | +12.3 (1.3) | +13.5 (0.6) | +16.6 (2.3) |
| Jua EPT-2 HRRR     | +12.0 (0.7) | +12.8 (3.3) | +14.9 (1.4) | +9.2 (0.4)  | +8.7 (1.3)  | +17.2 (3.5) |
| DWD ICON-EU        | +12.4 (0.8) | +7.7 (1.1)  | +11.8 (2.3) | +10.6 (1.1) | +11.3 (1.5) | +17.2 (3.2) |

**Table S18:** Track B MAE skill (%), 1–48 h – Figure 5, solar.

| Model              | All         | <P5         | P5–25       | P25–75      | P75–95      | >P95        |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Jua EPT-2.1 Helios | +12.6 (2.5) | +21.6 (4.6) | +18.1 (5.5) | -11.3 (0.5) | +13.5 (5.8) | +32.2 (4.9) |
| Jua EPT-2.1 Europa | +7.8 (2.0)  | +7.4 (3.9)  | +12.8 (3.1) | +10.7 (1.7) | +0.1 (4.5)  | +3.4 (7.8)  |
| Jua EPT-2 HRRR     | +0.8 (2.2)  | -0.2 (5.1)  | +4.8 (5.7)  | +6.5 (0.3)  | -6.1 (8.2)  | -6.4 (7.7)  |
| DWD ICON-EU        | +7.2 (1.4)  | +12.4 (3.3) | +12.4 (2.5) | +1.0 (2.2)  | +0.9 (3.3)  | +10.5 (5.7) |

**Table S19:** Track B MAE skill (%) by lead – Figure S9, wind.

| Model          | 6 h  |       | 12 h |       | 24 h |       | 48 h |       |
|----------------|------|-------|------|-------|------|-------|------|-------|
|                | All  | >P95  | All  | >P95  | All  | >P95  | All  | >P95  |
| EPT-2.1 Europa | +8.6 | +11.1 | +9.3 | +12.0 | +8.9 | +11.9 | +8.4 | +12.3 |
| EPT-2 HRRR     | +2.3 | +3.8  | +3.3 | +5.0  | +3.9 | +5.4  | +4.7 | +6.7  |
| ICON-EU        | +7.5 | +14.6 | +7.7 | +14.7 | +7.1 | +13.9 | +5.4 | +13.7 |

**Table S20:** Track B MAE skill (%) by lead – Figure S9, temperature.

| Model          | 6 h   |       | 12 h  |       | 24 h  |       | 48 h  |       |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|
|                | All   | >P95  | All   | >P95  | All   | >P95  | All   | >P95  |
| EPT-2.1 Europa | +10.7 | +12.7 | +14.6 | +17.0 | +13.2 | +15.1 | +13.5 | +16.1 |
| EPT-2 HRRR     | +12.0 | +17.1 | +14.2 | +19.3 | +12.4 | +16.8 | +12.2 | +16.9 |
| ICON-EU        | +13.7 | +18.4 | +14.9 | +19.9 | +13.2 | +17.2 | +10.6 | +15.5 |

**Table S21:** Track B MAE skill (%) by lead – Figure S9, solar (Ovc. = overcast, P5–P25).

| Model          | 6 h   |       | 12 h  |       | 24 h  |       | 48 h  |       |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|
|                | All   | Ovc.  | All   | Ovc.  | All   | Ovc.  | All   | Ovc.  |
| EPT-2.1 Helios | +12.9 | +20.3 | +11.4 | +17.1 | +10.4 | +16.4 | +11.2 | +16.8 |
| EPT-2.1 Europa | +7.2  | +13.6 | +8.0  | +13.5 | +6.8  | +12.3 | +6.9  | +11.9 |
| EPT-2 HRRR     | -3.5  | +3.9  | -1.5  | +4.7  | -1.3  | +4.6  | +1.1  | +6.4  |
| ICON-EU        | +7.4  | +14.1 | +7.8  | +13.4 | +6.0  | +12.0 | +4.4  | +9.3  |

**Table S22:** Track B precipitation MAE skill (%) by lead (Figure S11). Point estimates.

| Model              | 6 h  |      | 12 h |      | 24 h |      | 48 h |      |
|--------------------|------|------|------|------|------|------|------|------|
|                    | All  | >P95 | All  | >P95 | All  | >P95 | All  | >P95 |
| Jua EPT-2.1 Europa | +1.9 | +0.5 | +9.9 | +1.0 | +9.0 | +1.6 | +7.7 | +1.9 |
| Jua EPT-2 HRRR     | -2.5 | -0.6 | -2.6 | +0.6 | -0.9 | +1.0 | +0.4 | +1.7 |
| DWD ICON-EU        | -0.1 | +0.4 | +8.7 | -0.6 | +7.1 | -0.1 | +3.2 | -1.2 |

**Table S23:** Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure S10. Parentheses: jackknife SE.

| Model              | All                | Dry                 | <P50                | P50–75              | P75–95              | >P95               |
|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
| Jua EPT-2.1 Europa | +7.8 ( $\pm 0.9$ ) | +29.3 ( $\pm 1.4$ ) | -19.1 ( $\pm 6.0$ ) | -20.2 ( $\pm 7.2$ ) | -9.0 ( $\pm 5.7$ )  | +1.2 ( $\pm 0.8$ ) |
| Jua EPT-2 HRRR     | +1.4 ( $\pm 3.4$ ) | -3.6 ( $\pm 9.7$ )  | +10.3 ( $\pm 4.4$ ) | +15.6 ( $\pm 6.1$ ) | +10.4 ( $\pm 2.5$ ) | -0.7 ( $\pm 0.3$ ) |
| DWD ICON-EU        | +3.5 ( $\pm 0.5$ ) | +15.3 ( $\pm 0.3$ ) | -10.9 ( $\pm 2.7$ ) | -11.4 ( $\pm 3.8$ ) | -6.9 ( $\pm 2.2$ )  | +0.1 ( $\pm 0.5$ ) |

**Table S24:** Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure S7. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

| Model               | POD  | FAR  | CSI  | Freq. bias |
|---------------------|------|------|------|------------|
| Jua EPT-2.1 Europa  | 0.28 | 0.62 | 0.19 | 0.74       |
| DWD ICON Global     | 0.29 | 0.65 | 0.19 | 0.82       |
| Jua EPT-2 Reasoning | 0.23 | 0.56 | 0.18 | 0.53       |
| ECMWF IFS           | 0.25 | 0.65 | 0.17 | 0.71       |
| NOAA GFS            | 0.25 | 0.66 | 0.17 | 0.74       |
| Jua EPT-2 HRRR      | 0.20 | 0.53 | 0.16 | 0.42       |
| Jua EPT-2e          | 0.20 | 0.55 | 0.16 | 0.44       |
| ECMWF ENS (mean)    | 0.15 | 0.50 | 0.13 | 0.29       |

## A.5 COUNTRY-LEVEL REGIME PROFILES

Figures S12, S13, S14, and S15 show how the March–June regime-skill patterns vary across countries for the mixed regional/global comparison set. Countries without sufficient matched observations for a variable are omitted.

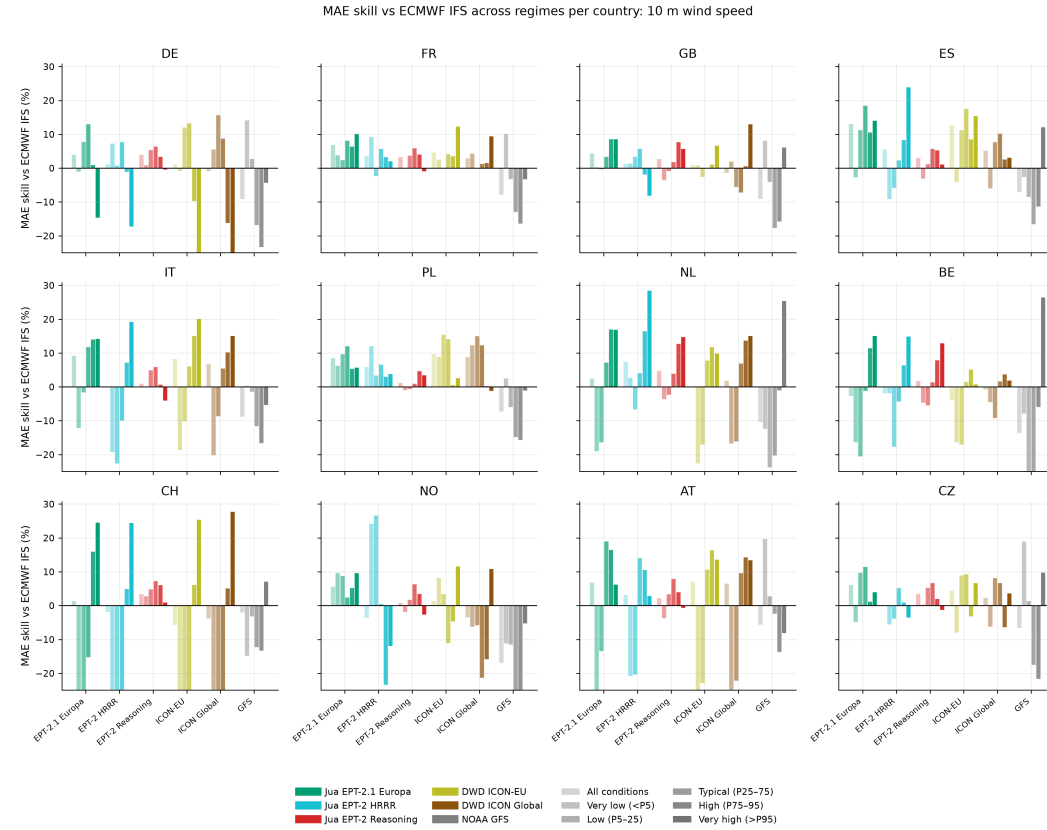


Figure S12: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set.

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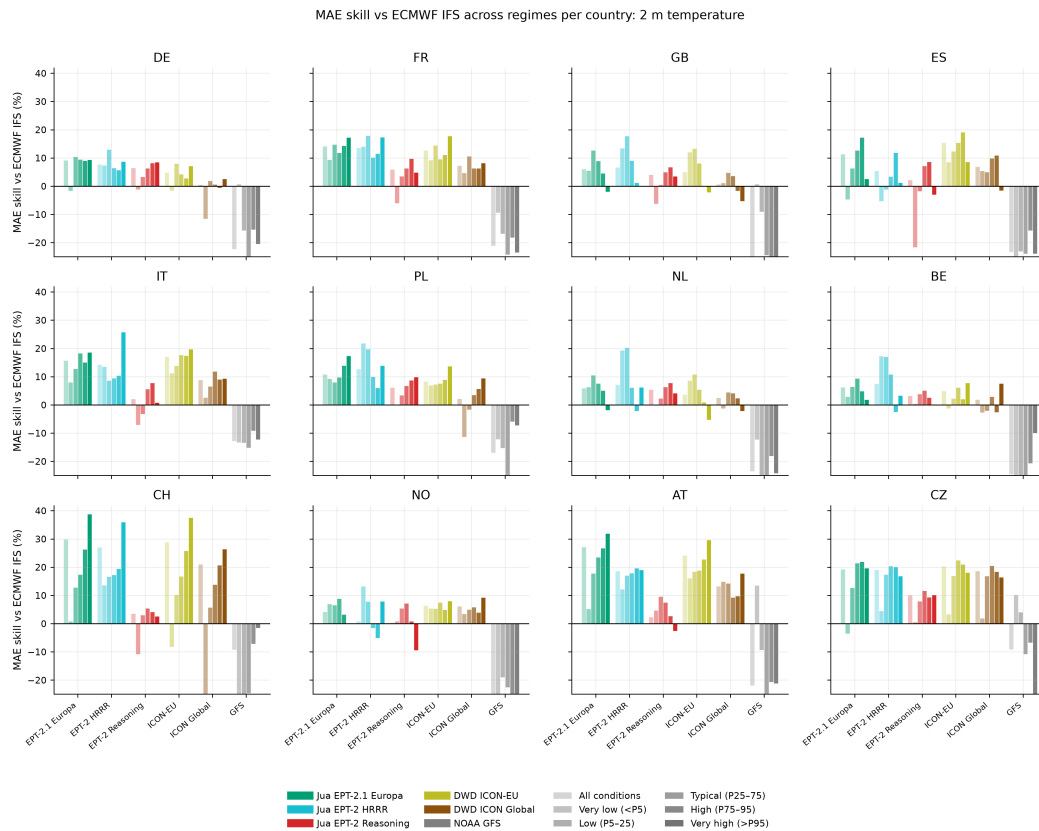


Figure S13: As Figure S12, for 2 m temperature.

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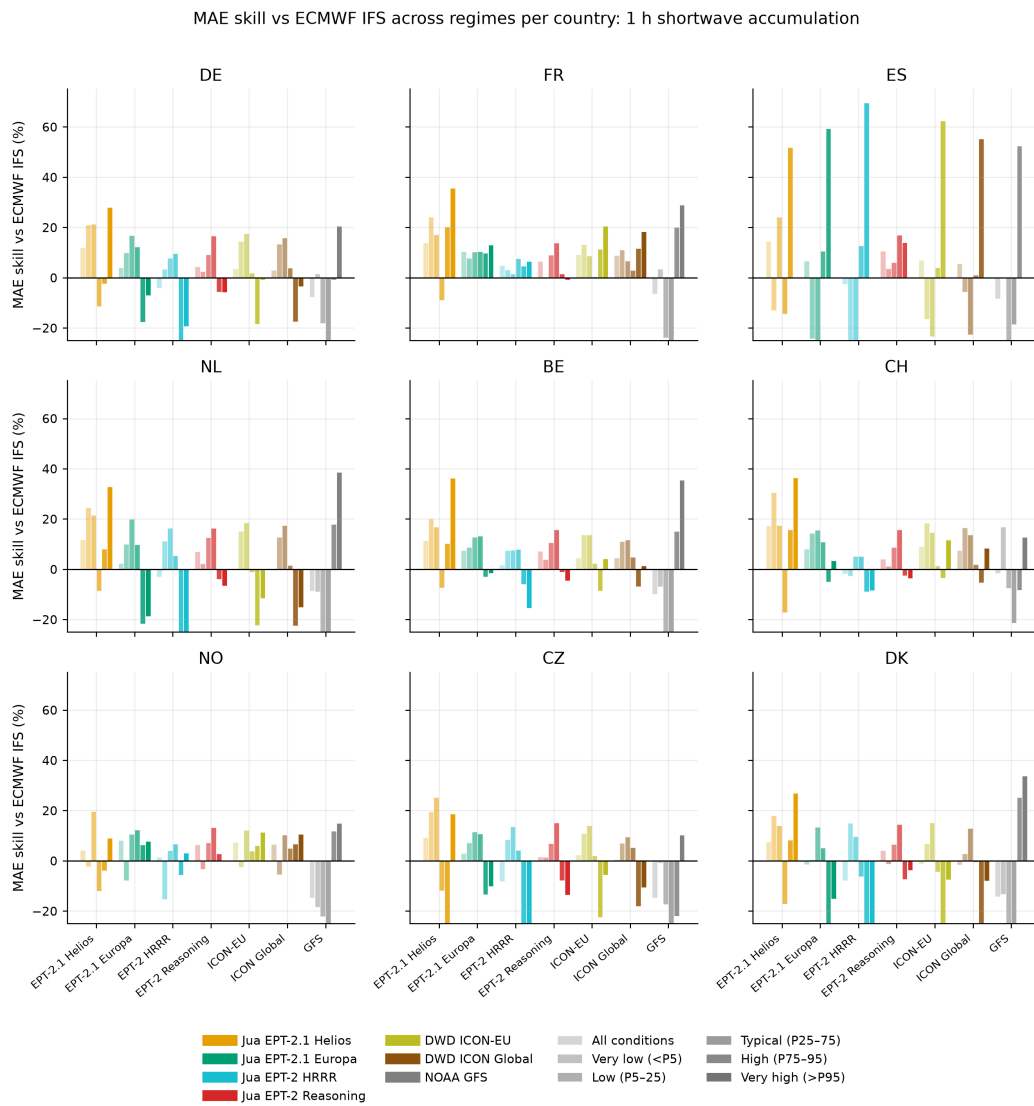


Figure S14: As Figure S12, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 2.2).

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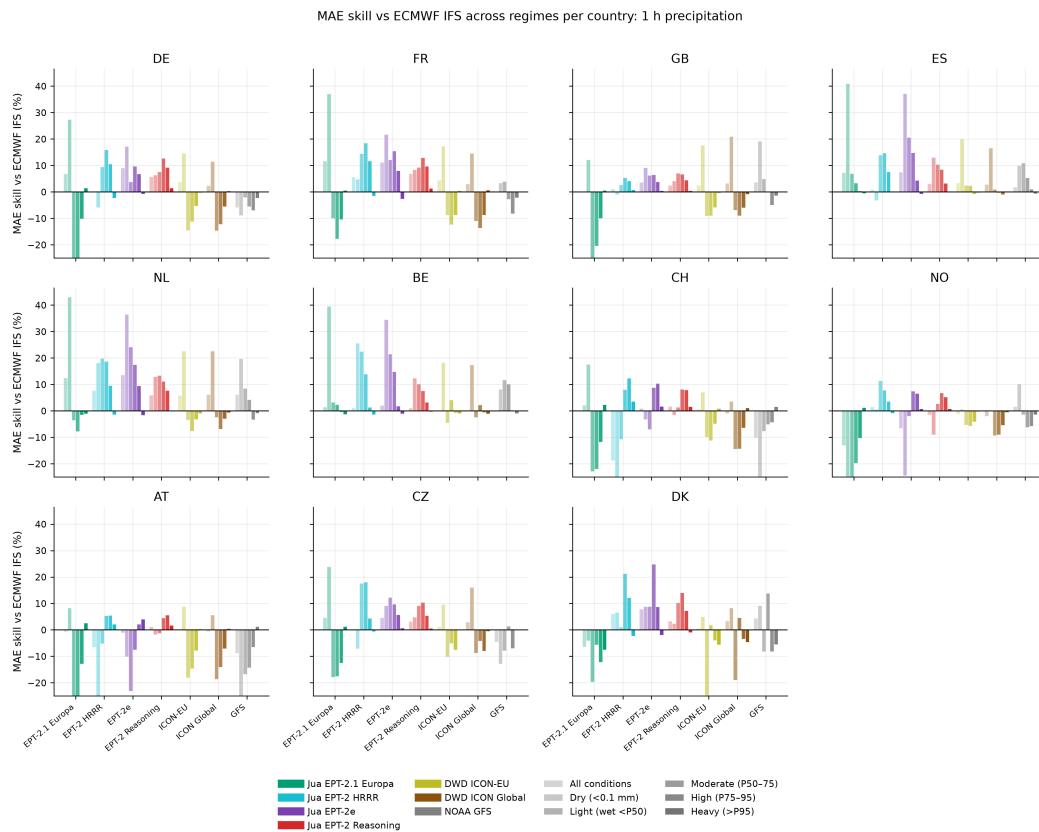


Figure S15: Per-country MAE skill by precipitation regime for the mixed regional/global model set. IT and PL have no gauges in the precipitation panel.

## A.6 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

Table S25 lists the externally documented temperature-record events and European windstorms used in the case-study analysis. Event windows and affected countries were fixed from the source catalogue before scoring. Every benchmark station in the source-reported countries is included at 00, 06, 12, and 18 UTC throughout the event window. The analysis is therefore conditioned on documented events, rather than on whether an individual station observation breaks a record.

For event  $e$  and lead  $\ell \in \{6, 12, \dots, 48\}$  h, let  $\mathcal{C}_{e,\ell}$  denote the station–valid-time pairs  $(s, t)$  in the event footprint for which ECMWF IFS and every Track A model included in the comparison provide a forecast initialised at  $t - \ell$ . Thus, each model is evaluated against the same observations at each lead. For ensemble models,  $f_m$  denotes the ensemble mean and  $\tilde{f}_m$  the forecast after the causal bias correction of Section 2.4. We compute

$$\begin{aligned}\varepsilon_m(s, t, \ell) &= \tilde{f}_m(s, t | t - \ell) - y(s, t), \\ N &= \sum_e \sum_\ell |\mathcal{C}_{e,\ell}|, \\ \text{MAE}_m^{\text{event}} &= \frac{1}{N} \sum_e \sum_\ell \sum_{(s,t) \in \mathcal{C}_{e,\ell}} |\varepsilon_m(s, t, \ell)|, \\ S_m^{\text{event}} &= 100 \left( 1 - \frac{\text{MAE}_m^{\text{event}}}{\text{MAE}_{\text{IFS}}^{\text{event}}} \right).\end{aligned}\tag{5}$$

The same observation may contribute at multiple leads because these are distinct forecasts from different initialisation times. For uncertainty, we delete one complete event at a time and report the jackknife standard error

$$\sqrt{\frac{E-1}{E} \sum_e \left( S_{m,(-e)}^{\text{event}} - \bar{S}_{m,(-\cdot)}^{\text{event}} \right)^2},$$

where  $E = 5$  for temperature and  $E = 7$  for wind. Complete pooled values and event-level uncertainty are reported in Table S26.

**Table S25:** Source-defined event catalogue. The accompanying reproduction package records the source URLs and selection basis for every event.

| Variable    | Event                                   | Dates                 | Affected countries                     |
|-------------|-----------------------------------------|-----------------------|----------------------------------------|
| Temperature | Central European December record warmth | 2025-12-07–2025-12-09 | NL, CZ                                 |
| Temperature | Late-February European record warmth    | 2026-02-25–2026-02-27 | FR, DE, ES, CZ                         |
| Temperature | Early-April record warm spell           | 2026-04-05–2026-04-08 | FR, GB                                 |
| Temperature | Western European May heatwave           | 2026-05-21–2026-05-30 | FR, GB                                 |
| Temperature | Late-June European record heatwave      | 2026-06-21–2026-06-28 | DE, FR, GB, ES, PL, NL, CH, AT, CZ, DK |
| Wind        | North Sea autumn storm                  | 2025-09-15–2025-09-16 | NL                                     |
| Wind        | Storm Amy                               | 2025-10-03–2025-10-05 | GB, FR, BE, NL, NO, DK, PL             |
| Wind        | Storm Benjamin                          | 2025-10-23–2025-10-24 | FR, NL, CH, GB                         |
| Wind        | Storm Goretti                           | 2026-01-08–2026-01-09 | GB, FR, DE, NL                         |
| Wind        | Storm Nils                              | 2026-02-11–2026-02-13 | FR, ES                                 |
| Wind        | Storm Marlis                            | 2026-03-25–2026-03-26 | CH, DE                                 |
| Wind        | Storm Deborah                           | 2026-03-26–2026-03-27 | FR, IT                                 |

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**Table S26:** Event-case MAE skill (%) relative to ECMWF IFS. Parentheses give leave-one-event-out jackknife SE. Temperature uses five events and 231,065 common samples; wind uses seven events and 123,847 common samples.

| Model                 | Temperature events | Windstorms  |
|-----------------------|--------------------|-------------|
| Jua EPT-2.1 Europa    | +15.4 (1.6)        | +6.4 (1.8)  |
| Jua EPT-2 HRRR        | +12.3 (2.7)        | +3.9 (1.3)  |
| Jua EPT-2 Reasoning   | +6.5 (0.6)         | +3.3 (0.5)  |
| Jua EPT-2e            | +4.3 (1.1)         | -0.3 (0.8)  |
| Microsoft Aurora      | +2.5 (0.9)         | +0.7 (0.9)  |
| ECMWF AIFS            | -3.4 (3.9)         | -6.0 (1.3)  |
| ECMWF AIFS ENS (mean) | +0.1 (2.0)         | -5.1 (0.7)  |
| ECMWF ENS (mean)      | -0.9 (0.1)         | +1.7 (0.7)  |
| NOAA GFS              | -25.4 (7.5)        | -12.6 (1.8) |
| DWD ICON Global       | +9.8 (2.3)         | +2.4 (2.2)  |